17.19. $\theta = 1.22\cdot\frac{\lambda}{D} = \frac{\Delta d}{x}$

$$x = \frac{\Delta d\cdot D}{1.22\cdot\lambda} = \frac{3.0mm\times 1.5m}{1.22\times 550nm} = 6.7km.$$

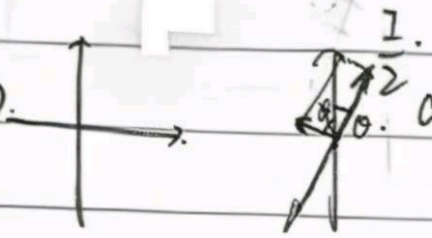
17.21. $2\times d\cdot\times\frac{\sqrt{2}}{2} = k\cdot\lambda\cdot,\ k\in Z$

$$\lambda = \frac{\sqrt{2}\times 0.275nm}{k} = \frac{0.389}{k}.$$

$k=3$ 时，$\lambda = 0.1297\cdot nm \approx 0.13nm$

$k=4$ 时，$\lambda = 0.0975\ nm. \approx 0.097nm.$

18.2. (1).



$\cos\theta = \sqrt{\frac{\frac{I}{2\times 3}}{\frac{I}{2}}} = \frac{1}{3}$，$\theta = \arccos\frac{1}{3}$

$= 54°44'$

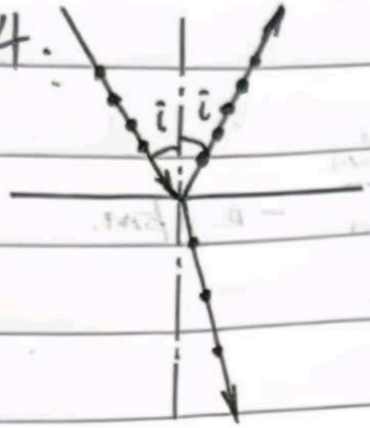
(2). $\cos\theta = \sqrt{\frac{\frac{I}{3}}{\frac{I}{2}}} = \frac{\sqrt{6}}{3}$，$\theta = \arccos\frac{\sqrt{6}}{3} = 35°16'$

18.3. $I_1 + I_2\times\frac{1}{2} = 5(0 + I_2\times\frac{1}{2})$

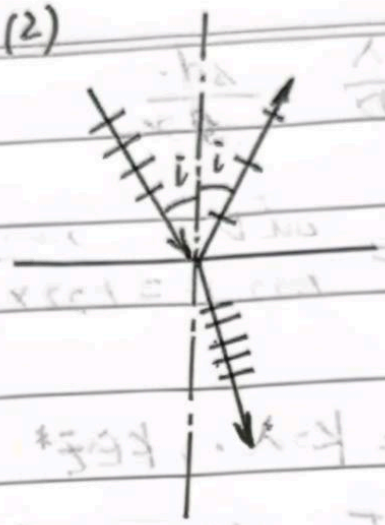
则 $\frac{I_1}{I_2} = 2$. 则线偏振光与自然光光强之比为2:1.

18.4.

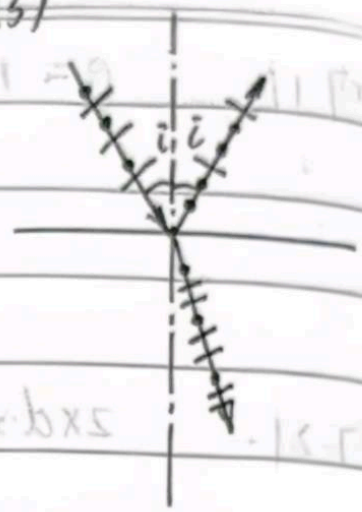
(1)



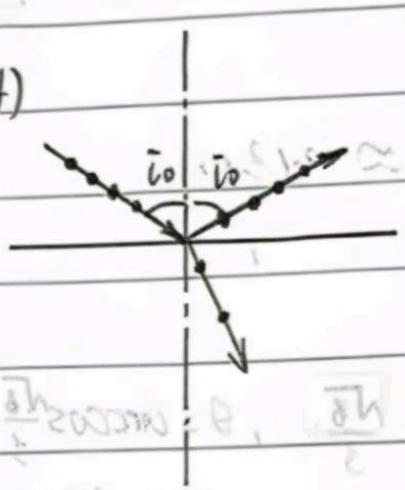
(2)



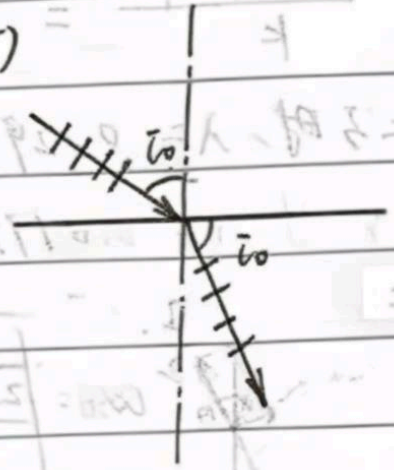
(3)



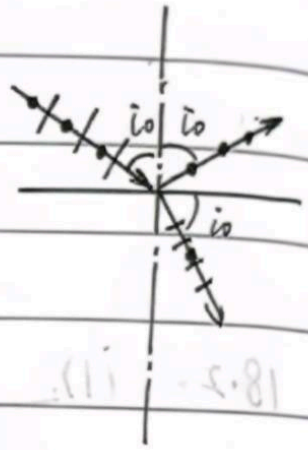
(4)



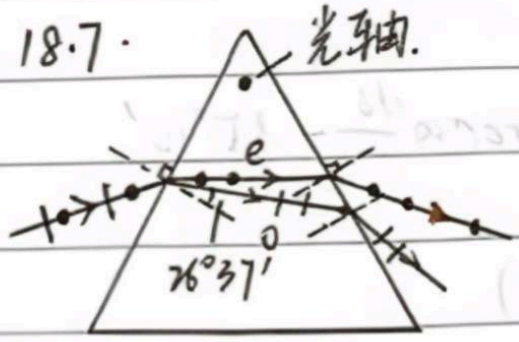
(5)



(6)



18.7.



$$\frac{\sin i}{\sin(30^\circ)} = 1.486.$$

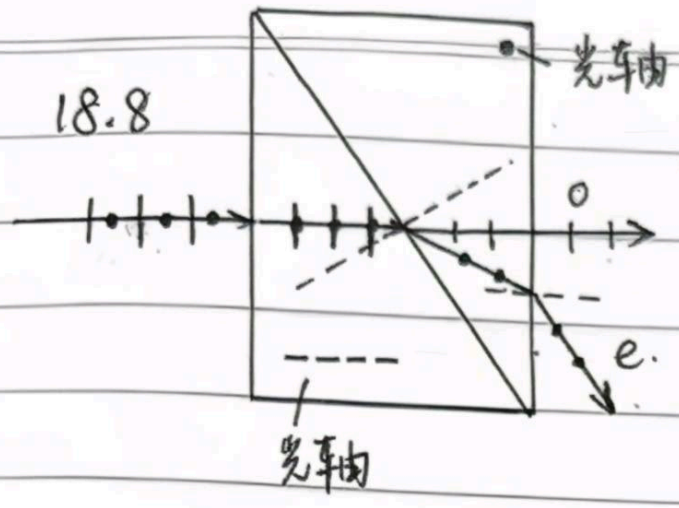
则 $i = 47^\circ 59'$

入射角为 $47^\circ 59'$

$$\frac{\sin i}{\sin\theta_0} = n_0 = 1.658$$

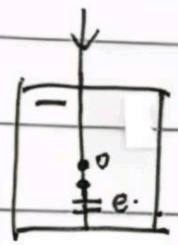
则 $\theta_0 = 26^\circ 37'$

18.8



18.9. $d_{(\frac{1}{4})} = \frac{\lambda}{4(|n_o - n_e|)} = \frac{589.3nm}{4\times(1.553-1.541)} = 1.23\times10^{-5}m \approx 0.012mm.$

18.10.



(1). $I_{寻} = \left(\frac{1}{2}\right)^2 I_0 = \frac{1}{4} I_0.$

$I_{非} = \left(\frac{\sqrt{3}}{2}\right)^2 I_0 = \frac{3}{4} I_0.$

(2). $\Delta\varphi = \frac{\pi}{2}.$ $\Delta\delta = \frac{\lambda}{4}.$

$d = \frac{\lambda}{4|n_o - n_e|} = 8.6\times10^{-7}m$

18.11. (1). $\delta = (n_o - n_e)d.$

$\Delta\varphi = \frac{2\pi}{\lambda}(n_o - n_e)d = 2k\pi., k\in Z.$

则. $d = \frac{\lambda}{(n_o - n_e)} = \frac{400nm}{1.658-1.486}$

$= 2.33\times10^{-6}m.$

(2). $\Delta\varphi = (2k+1)\pi.$ $d = \frac{\lambda}{2(n_o - n_e)}$

$= 1.16\times10^{-6}m.$

18.14. $I_0 = 2I$